

1. trigger and stored procedure

Aspect	Trigger	stored procedure
definition	Special stored procedure that runs automatically when various events happen (insert, update, delete).	pieces of the code written in to perform some specific tasks.
Execution	Automatic	Manual
Calling	Cannot be called	Can be called
parameters	Dose not accept parameters	accept paramerters
Dependency	Must be attached to table	Independence of tables

2. stored procedure and functions

Stored procedure	functions
Can return zero, a single value, or several values.	Returns a single value, either as a table or as a scalar,
The database contains stored procedures that have been parsed and compiled.	Run-time compilation and execution occur for functions.
Support input and output parameter	output parameter not supported
Allow the use try and catch	Cannot use try and catch

3. drop and delete statement

Drop	Delete
DDL (data definition language)	DML (data Manipulation Language)
Removes data and structure	Removes data only
Not allowed to use WHERE clause	Allowed to use WHERE clause to delete specific rows
Dose not fire triggers	Fire triggers
Cannot rollback	Can rollback
Faster	Slower

4. select and select into statement

select	select into
Display Data	copies data into new table
Does not create table	Creates new table automatically
Destination table must exist	Destination table must not exist
Dose not change database structure	Column data types inherited from original table

5. DDL, DML, DCL and DQL

DDL	DML	DCL	DQL
Data definition language	Data manipulating language	Data control language	Data querying language
define, create, modify, or remove database objects	used to manipulate data inside tables	control access and permissions on database objects	retrieve data only
Works on database structure	Works on table data	Manage security	Dose not modify data or structure
CREATE, ALTER , DROP, TRUNCATE	INSERT, UPDATE, DELETE, MERGE	GRANT, REVOKE, DENY	SELECT

6. Table valued and multi statement function

Table valued	multi statement
Single SELECT	Multiple statements
Dose not return table variable	return table variable
Faster	slower

7. Varchar(50) and varchar(max)

Varchar(50)	varchar(max)
Variable-length string with a maximum of 50 characters.	Variable-length string with a maximum of $2^{31}-1$
Good for short text fields	Used for very long text data like xml

8. SQL and windows Authentication

SQL Authentication	windows Authentication
SQL Server uses its own login system	SQL Server uses Windows user accounts to log in
Can be used for non-Windows users	Authentication is handled by Windows
SQL server users cannot access windows files if have no permission	SQL server users can access windows files

9. Inline function and view

Inline function	view
Accept parameters	Cannot accept parameters
Parameterized view	Stored select statement
Can not be indexed	Can be indexed

10.Identity and unique Constraint

Identity	unique Constraint
Auto generate non repeated numbers	Ensures no duplicate values
Column property	constraint
Only one IDENTITY column per tabl	Can be applied to one or multiple columns
	Allow one null